

Persuasive Speech Transcript

Hello everybody! My name is Garrett Brownlee, and today I want to bring a serious issue to your attention: The regulation of "dopamine-hacking," which is the abuse of biological mechanisms by companies and software engineers alike.

A. Attention Getter: (get your audience interested in your speech):

Testimonies from our fellow Speech classmates perfectly highlight how this "brain-hijacking" struggle affects their lives. Rosalynd Van Pelt, said this: "When I had difficult papers to write, I lost motivation and used scrolling as a way to forget it. Even if the app had an automated suggested break, my attention was already too low to consider my present moment." Another classmate of ours was kind enough to share their own personal impact, saying, "While I can't recall a specific instance of this, I know it has happened to me multiple times. It makes me so sad to think that I am not being aware or hospitable to people that I care about when I'm on my phone!"

B. Relevance: (state why this topic and information is relevant and valuable to your audience):

I'm pretty confident that each person listening can relate to these testimonies on some level, and that is exactly what makes this such an important topic! This issue affects the majority of people across the world every single day. The issue has an even greater importance for students, since technology and software are entirely intertwined with education these days. Beyond that, with all of us being in our twenties, we are all the "target audience" of social media companies, online shopping, and streaming services alike. We can all relate to that feeling of "brain fry" after scrolling too long on Instagram or bingeing Netflix for hours at a time.

C. Credibility: (state your interest why you are qualified to talk about this topic):

As a neuroscience major, I have studied the specific pathways and structures involved in the brain's dopamine "reward system" for the last four years. Each year, this topic becomes more relevant to my coursework as the issue's prevalence and severity grow increasingly. Here are some of these courses I'm describing, which I have taken in my time as an undergraduate student at Baylor University: NSC 3311 (Cognition), NSC 3320 (Learning & Behavior), NSC 3370 (Affective), and NSC 4330 (Advanced Principles). In all four of these courses, we discussed the "reward pathway" and all the relevant structures, which we call the "mesolimbic dopamine pathway" in neuroscience. In just a second, I will dive deeper into these "reward" structures and neuro-biological mechanisms behind "dopamine-hijacking."

Transition: *This brings me to the 3 main points of today's speech*

D. Thesis: (state your goal/purpose):

Because technology companies intentionally exploit our neurobiology to keep us hooked, we must recognize that self-control is insufficient, and thus advocate for more neuro-ethical design standards (at the corporate level) and legislative policies (at the governmental level) to restore our cognitive autonomy.

E. Preview: (state your plan for the body; *In this speech I am going to...*):

In this speech, I will examine the science of neurobiological exploitation, explain why willpower alone fails, and provide a policy-based solution to reclaim control over our attention.

Transition: *To begin, we must look at the actual neuro-biological mechanisms that developers exploit to hijack our attention.*

So what exactly is "dopamine-hijacking," and how does it disrupt one's life?

Main Point #1: There are numerous mechanisms behind how technology companies and software developers intentionally exploit the dopamine systems of our neurobiology, which keep us engaged with their product or provided service. Two keep it short and simple, I'll focus on the primary two methods.

- a) **Subpoint:** One primary method of “dopamine-hijacking” is utilizing it to lock users into an “infinite scroll” (better known to some of you as the “doom scroll”) so that their product becomes your primary focus, and you forget all your pressing responsibilities
 - a. **Sub-Subpoint:** “One technique used to prolong usage time in this context is the endless scrolling/streaming feature. For example, on the video platform YouTube one can scroll down endlessly with apparently no end. By endlessly scrolling down, the user is getting more and more immersed (perhaps also on a motor level as perceiving the scrolling as a playful activity) while not coming to a natural stop, where he/she might easily reconsider to quit or change the platform. The behavior of endless scrolling is enhanced, because the user finds from time to time something rewarding (e.g., a funny or interesting video), hence intermittent conditioning principles are observed here. The same phenomenon is at work when ‘auto play’ is the default” (Montag et al., 2019). This design choice prevents the brain from processing the "completion" of a task, leading to the "brain fry" many of us feel.
 - b. **Sub-Subpoint:** “Flow itself is a positive state of mind, which can promote high productivity...however, it can actually be ‘fatal’ in the case of applications on the smartphone or other ‘ubiquitous’ devices, when attention should be focused somewhere else: It is well known that flow goes along with a feeling of time distortion and this is exactly what many developers of social media apps and Freemium games aim to achieve—a person being so immersed that he or she is forgetting about time and space while using a platform or app” (Montag et al., 2019).
- b) **Subpoint:** Another primary method of dopamine-hijacking is the use of variable-ratio reinforcement.
 - a. **Sub-Subpoint:** “Most dopamine neurons in the midbrain of humans, monkeys, and rodents signal a reward prediction error; they are activated by more reward than predicted (positive prediction error), remain at baseline activity for fully predicted rewards, and show depressed activity with less reward than predicted (negative prediction error)” (Schultz, 2016).
 - b. **Sub-Subpoint:**
- c) **Subpoint:** Dopamine-hijacking’s impact on student well-being.
 - a. **Sub-Subpoint:** “A meta-analysis of several dozen research studies indicates that higher levels of switching between different media channels is significantly linked to lower levels of both working memory and long-term memory. Given the current Extractive Attention Economy, and the increasing number of social media platforms and apps competing to capture our attention, basic human capacities — such as our memories — are increasingly under attack” (Center for Humane Technology, 2018).
 - b. **Sub-Subpoint:** “A long-term study of several hundred Dutch teenagers shows that problematic social media use is significantly linked to the emergence of serious cognitive effects a year later, including reduced attention, increased impulsivity, and increased hyperactivity” (Center for Humane Technology, 2018).
 - c. **Sub-Subpoint:** My fellow classmates are essentially the "victims" of these strategies, facing daily battles with their own focus.

Transition Sentence: *While many believe they can simply "turn it off," the biology of the brain proves that personal responsibility is not a fair fight against intentional, exploitative engineering.*

II. Main Point #2: We must shift from "self-control" to policy-based solutions

A. Subpoint: We must address the counterargument that distraction is merely a "failure of willpower."

1. Sub-Subpoint: A common belief is that distraction is a personal failure, but neuroimaging evidence shows that these apps actually suppress the prefrontal cortex (Lorenz, R. C., et al. (2015)).
2. Sub-Subpoint: Because this area of the brain is required for executive function, users are biologically disadvantaged when trying to resist.

B. Subpoint: Benefits and procedure for implementing neuro-ethical design standards.

1. Sub-Subpoint: We must support legislative policies, like the SMART Act, to limit exploitative features used in software engineering (SMART Act, S.2314, 116th Cong. (2019)). The 116th Congress was in session from January 2019 to January 2021.

C. Subpoint: Shifting the burden of responsibility from the individual onto the policy-makers and developers

1. Sub-Subpoint: My goal is to pursue a career where I serve people and protect their health; this starts with advocating for neuro-ethics (Bhargava, V. R., & Velasquez, M. (2021)).
2. Sub-Subpoint: Policy restores cognitive autonomy by ensuring companies can no longer operate outside the confines of law.

Transition Sentence: (signal to the audience you have finished Main Point #2; introduce Main Point #3):
Once we have protective policies in place, we can visualize a healthier relationship with our devices.

III. Main Point #3: What life looks like when we reclaim our cognitive autonomy.

A. Subpoint: Imagine what life could be like with you back in the "driver's seat."

1. Sub-Subpoint: Imagine the mental clarity you would gain by being in complete control of your own mind and actions.

2. Sub-Subpoint: This clarity allows for greater academic success and more meaningful professional relationships.

B. Subpoint: Let's hear an inspiring transformation story from somebody who has reformed their technology use.

1. Sub-Subpoint: Consider a student who, after breaking free from these dopamine loops, regained their focus and peace of mind.
2. Sub-Subpoint: This transformation shows that we deserve to be in control of our own cognitive health.

C. Subpoint: Cognitive autonomy will help protect your future careers and well-being.

1. Sub-Subpoint: As future professionals, we must ensure our focus is preserved to be successful in our chosen fields.
2. Sub-Subpoint: Reclaiming our autonomy means technology becomes a tool we use, rather than a system that uses us.

Transitional Phrase: (signal to audience you have finished all main points and will now conclude the speech): We have analyzed the science and visualized a healthier “online” future; now let’s review what we’ve discussed and let’s really emphasize what it is that we all need to do to resolve this issue.

Conclusion

- I. **Summary:** (Restate goal and main points): We have examined the science of exploitation, why willpower is insufficient against engineering, and how the SMART Act can restore our autonomy. You can take action yourself by supporting neuro-ethical legislation and advocating for more standards that protect our collective cognitive health. Give your local congressmen a call, they’ll never make a change if they don’t know the extent of this issues negative affects.
- II. **Memorable Close:** (Provide closure and leave a lasting impression): To conclude, I want to leave you with a comparison: casinos are designed to turn off executive-functioning regions of your brain through the intentional lack of windows and clocks to keep you gambling. Technology should be a tool that we pick up and put down, not a casino with no clocks. We must not let ourselves become enslaved by it, forgetting our responsibilities...forgetting ourselves. Let’s all advocate for a world where the exit door is easy to find and reclaim control over our lives!

References

Bhargava, V. R., & Velasquez, M. (2021). Ethics of the Attention Economy: The Problem of Social Media Addiction. *Business Ethics Quarterly*, 31(3), 321-359.

Center for Humane Technology. (2023). *Ledger of Harms: Dopamine and the Brain*.

- Lorenz, R. C., et al. (2015). Video game training and the prefrontal cortex. *Molecular Psychiatry*, 20, 34–39.
- Montag, C., Lachmann, B., Herrlich, M., & Zweig, K. (2019). Addictive features of social media/messenger platforms and free-to-play games: A design perspective from psychology and informatics. *International Journal of Environmental Research and Public Health*, 16(14), 2612.
- Schultz, W. (2016). "Dopamine reward prediction error coding." *Dialogues in Clinical Neuroscience*.
- Social Media Addiction Reduction Technology (SMART) Act, S.2314, 116th Cong. (2019).